

RoomMe: Matching Roommates Based On Suitability With Deep Neural Networks and Evolutionary Algorithms*

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Abstract. This report presents a method for matching roommates on a pair basis, using artificial neural networks, and one that builds on this technique, which finds a solution to a modified version of the stable matching problem, expressed in finding the most suitable combinations of roommates for a whole arbitrary list of candidates. Our neural network, when trained on pairs of roommates, outputs a fundamental for our problem approximation score of roommate suitability. It could be used stand-alone for pairing roommates or as a fitness value estimator for our evolutionary solution to the stable matching problem.

Keywords: roommate matching · deep learning · artificial intelligence
· evolutionary algorithm

1 Introduction

Artificial Intelligence in the modern world is used ubiquitously and scientists are looking for newer ways to apply it to advance our lives. Deep neural networks have proven to learn from big data sets and generalize well to new data points, if given sufficient training.

Motivated by the accuracy and usability of AI techniques, we propose a novel way to match roommates, based on a artificial neural network (ANN), used as a suitability score estimator. We also broaden the usage of our ANN through an evolutionary algorithm to solve an altered version of the stable matching problem, adapted for the task of finding the most suitable groupings of roommates in a whole list of people. Our technique, which uses as basis a supervised dataset of people's habits and personality traits to train our ANN on labels of suitability scores, is a novel way of dealing with the problem of roommate matching through deep learning. It learns to approximate a suitability metric directly from people's

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features, as opposed to other works on the topic, where some hand-calculation is exercised.

By using these two methods, we cover virtually every case where roommates need to be matched - the single match case, where one person has to be matched to one or more roommates, and the list match case, where we have a list of people that need to be split into groups of n people that would be the best optimized for mutual suitability as roommates.

2 Problem

2.1 Domain

Students are faced with multiple challenges when having to settle in their university, one of which is finding where to live. Finding accommodation is a struggle for a lot of students, being students ourselves we know this from experience. The domain contains multiple problems, such as availability, suitability, price, etc. Housing organisations provide accommodation for students, thus resolving the availability problem, however, they have yet to find a solution for another main problem: suitability. Suitability of housing is an important problem in our domain, as having compatible roommates greatly influences the work-study experience. Being matched with non-compatible roommates decreases one's productivity, which is especially important for university students. On the contrary, suitable housemates with matching lifestyles, personal traits and professional interests create a collaborative and nurturing environment for the mind to thrive at its task at hand. Thus, the problem of suitability is a viable problem in the domain of housing. This motivates the purpose of our study.

In the problem of suitability we have identified a fundamental problem and a wider problem that we propose a solution for in our research:

2.2 Fundamental Problem

The first problem is assigning a suitability score between a pair of people. This is a fundamental block in our problem space, as it outputs an empirical measure of suitability. It could be used to directly match a person to one or more roommates, based on their personality traits and habits. It is also used in our larger-scale problem. The score that our neural network outputs is constrained within $[0; 1]$, because this is a normalized range of values, optimal for the learning of the ANN. It could then be scaled up to a more human-readable range, such as 0 to 100, where 0 means no suitability between a pair at all, and 100 - perfect suitability. That said, the ANN is just an approximator for suitability, so no 100 score means absolute perfection. What should be kept in mind is that there is always error with machine learning approximators, but if the trained model is accurate enough, the error is negligible, so it could be ignored.

2.3 Wider Problem

The wider problem we solve is finding the most suitable groups of roommates for a whole list of candidates to occupy a number of spaces in a residence using an evolutionary algorithm and our aforementioned neural network as the basis for calculating fitness value. In other words, if we have n number of candidates for n number of available accommodation spaces, we have to split the candidates into equal groups of r students, who share a common space, i.e. they are roommates in any kind of sense. r could be an arbitrary number less than n . There should be another empirical measure of suitability of an arrangement (permutation) of a list of candidates, put into groups. Even if we do not have even groups of students to be roommates, we should still be able to assess a suitability score to any arrangement of candidates.

The upside of our approach is that we have the smallest possible unit of measurement of suitability, i.e. the ANN, which approximates suitability between a *pair* of roommates. This means that if we want to calculate the score of more than two people together, we can do so by calculating the score of each pair in the group of potential roommates and then calculate the composite score as a function of them. For example, if we have a set of people M number of potential roommates, we find all the combinations of two people (pairs) in M , approximate all pair scores and then plug them to a function. It could be a simple summation.

3 Related Work

3.1 Bethe Learning of Graphical Models via MAP Decoding

In this paper, a machine learning model is trained on a supervised dataset of students' answers to 14-question surveys of their habits and personalities. The labels are the returning students' picks for roommates. This is a different supervised approach from ours, as we use satisfaction (compatibility) scores, obtained through surveying the students in our dataset. Although a different dataset and technique are used, the input features of this paper are the ones we took inspiration from. Moreover, this study has found the most important traits of a roommate: smoking, personality and sleeping habits. These findings helped us construct the right features for our dataset.

3.2 Roommate Compatibility Detection Through Machine Learning Techniques

The study reviews different machine learning methods (Regression, Naïve Bayes, SVM) to perform analysis on the The Big Five personality traits dataset, which is a set of personal characteristics questions with answers ranging from 1-Negative through 5-Positive. Where their work differs from ours is in the fact that they perform an individual classification of "socialness" of a person. They hypothesise that the closer the socialness of two individuals, the bigger the chance that they will match as roommates. We disagree with this hypothesis, hence we use a direct metric of suitability.

3.3 Solving A Variation of The Stable Roommates Problem Using Evolutionary Algorithms

This paper proposes a genetic algorithm solution to a modified version of the stable matching problem, where instead of matching based on a list of preferred roommates for each person, they match students based on survey questions. That means they have a database of people's traits and have to find the best combinations of pairs of roommates in the whole database, i.e a solution to the stable roommates problem.

An evolutionary algorithm is used, where an environment of evolving individuals exists. Each individual has a genotype, which consists of a serialized permutation of pairs of people and their features. With each generation these individuals improve their fitness score, which in the case is a matching score. After some generations, the fittest individuals are picked as solutions to the problem.

The conclusion is that this method works in finding optimal stable pairings for a database of students. We were intrigued by this idea and decided to combine it with our ANN, which we use as a fitness score estimator. The authors of the study did not use any weighting for the features in the dataset in calculating the fitness score, but that might lead to sub-optimal results, as there is no sense of importance of the features. Moreover, even if they assigned weights to each one (as suggested in their Future Work section), this would still be a linear model, that would not take into account complex relationships between the different traits. For this reason, we use an ANN, which is capable of learning non-linear relationships in features (activation functions account for this quality of neural networks).

4 Expert Knowledge

As experts, we have chosen to speak to fellow students, as they are the target group for our project. We want to learn from their experience. What problems have they experienced in searching for a room? How could their search be optimized? What are the features they seek in a housing system and possible roommates? As noted earlier, instead of asking a few fellow students for a short interview, we chose a different approach. We made a survey about students' living situation and their experience. That way, we have reached even more students than we had hoped for.

Out of the 50 students that answered the survey, only eighteen live in a student facility. Out of the other 32 students, twelve students can't find a room in the city of their university and six students are still abroad and are not able to live in the Netherlands because of the current Covid-19 pandemic. Fourteen students have answered various different reasons for not living in student facilities. Most of them have said they have no desire to live there.

Most students with roommates have met them before accepting the room, but the same amount of students indicated that meeting them did not affect their

decision to move in. Furthermore, respondents want more roommates and/or are satisfied with their roommates. This means that collaborative living with other people is very important for students and gives motivation for the purpose of our study. We aim to match students with the most suitable available roommates for them, so that they could thrive in their environment through collaboration.

What further motivates our study is that 72,2% of the students would be interested in a compatibility system for finding roommates. The ones who aren't interested reason that such system might not function well.

When asked about qualities they wish to see in possible future roommates, cleanliness, honesty and respect (for boundaries) are high on the list. On the other hand, students dislike loudness, rudeness, messiness, smoking and drinking excessively in a possible roommate. These findings from our survey, along with the studies mentioned earlier, were a major influence in choosing the right features for our database.

Unfortunately we have not been able to get in contact with an employee from the housing agency DUWO so far, we hope that we will be able to interview one soon. We hope to get a new insight from the renter's point of view. How it might differ from the tenant's wants and needs for the platform. And is there a great difference between a good renter-tenant relationship and a good roommate-roommate relationship?

5 Methods

5.1 Dataset

The dataset we create is a labelled dataset, whose features are based on a standardized survey of 12 questions, assessing a person's personality, habits and other traits. This questionnaire uses a Likert scale, to yield empirical results, that we could use as input to our neural network after normalizing. The way we do that is by mapping the Likert scale answers to a range from 0 to 1 (0 being the most negative, 1 - the most positive answer). Table 1 shows the questions we used in our survey. They were picked based on the expert knowledge of our colleagues at VU about roommate compatibility and knowledge obtained from other studies [1–3]. They have had personal experience with student roommates, thus they provided valuable insight into what is the important data to be gathered.

The idea behind our database is inspired by the dataset of Tang, K et al. (2016) [2], where 14-question surveys that assess personality and habits are used as features. What we do differently is the labels we use: direct suitability scores between a pair of roommates, as opposed to picks for roommates of the people in the dataset, as in [2]. Moreover, the features we use are answers to our own survey.

As labels for the dataset we use the responses of students that have lived together for some time to record a happiness score on a scale from 0 to 100. This is done with the assumption that being happy with your roommate equals being compatible with this roommate. We then calculate the pair's combined happiness

Table 1. Table showing the questions in our survey for forming the features and the possible answers.

Questions	Answer format
Do you usually wake up early?	Liekert scale
Do you usually go to bed late?	Liekert scale
What is your gender?	0 for male, 1 for female
To what extent do you respect others' boundaries?	Liekert scale
How social are you?	Liekert scale
How regular of a smoker you are?	Liekert scale
How often do you drink and/or use drugs?	Liekert scale
How tidy of a person are you?	Liekert scale
Are you usually busy with work (university and jobs)?	Liekert scale
Do you get easily get angry?	Liekert scale
Are you an emotional person?	Liekert scale
To what extent are you loud and restless?	Liekert scale

score by averaging both answers' values. Then the score is normalized between 0 and 1. This way, we form pairings of student's personality assessments and a corresponding label of the student's experience with their particular roommate.

We use 3000 records of student pairs' answers to our survey and their corresponding happiness score answers to formulate a sufficient dataset for our neural network, as the number of combinations of features grows exponentially with linearly adding feature points. Moreover, neural networks are data-hungry and need a lot of data to generalise well.

5.2 Neural Network

Our fundamental algorithm consists of a neural network that maps the input features to the output labels. Ultimately, it learns to output a suitability score between a pair of roommates.

Let our neural network be denoted as $f(i; \theta)$, where i is a 24-dimensional input vector and θ are the optimizable parameters of the neural network, i.e weights and biases. Because the features for each person in our database are 12-dimensional vectors

$$v = [v_1, v_2, \dots, v_{12}] \quad (1)$$

and we input a pair of people's features to the ANN (v' and v''), our input vector is

$$i = [v'_1, v'_2, \dots, v'_{12}, v''_1, v''_2, \dots, v''_{12}], \quad (2)$$

which is why the input vector has 24 dimensions. The corresponding output is a scalar value

$$o = f(i; \theta), \quad (3)$$

bounded by $o \in [0; 1]$.

We chose a neural network for our task because we have a complex dataset of multiple features, which need a complex learning algorithm to be analysed.

The hidden layers of the neural network learn non-linear patterns in the training set, as opposed to linear models and most machine learning models.

The dictating of the weights of this neural network is done via the backpropagation algorithm with Adam [5] as an optimizer, which are by far the most widely used algorithms for training deep neural networks. Adam is an adaptive version of the classic stochastic gradient descent (SGD) [4] and is widely used as one of the most versatile optimizers. For this reason, we use it in our work.

Our trained model is used for approximating the suitability between a pair of roommates. That covers the case where two people have to be matched for roommates for a single house, or a roommate has to be matched to an existing resident. The best combination of people is sought through a set of candidates to find the best matched pair to accommodate a place.

5.3 Genetic algorithm¹ as a solution to the stable matching problem

The wider problem we solve is one where we have a list of n candidates, who have to occupy n places in a residence. In this case, we build on the idea of BERGER, R et al, (2018) [1], who use a genetic algorithm to build an environment, populated with individuals. These individuals' genes are a sequence of people (from the candidate list in our case), which we later divide into groups of r roommates (pairs, triplets, etc.) as a phenotypic expression. So, we define r as the group size.² Each person from the database should have answered our survey and we use the same data processing techniques as in section 5.1.

Each number in the genome corresponds to a person (index) in the dataset of people. For example, suppose we have an individual with genes $g = [2, 4, 1, 3, 5, 6]$. The first gene encodes the 2nd person into the genome, the second - the 4th person and so on. The genome also means that if $r = 2$, its phenotypic expression would be matching them the following way: $[\{2, 4\}, \{1, 3\}, \{5, 6\}]$. We could also form groups (not only pairs) of people per shared space, for example, groups of four ($r = 4$). Then, the individual in the evolutionary algorithm is encoded the same way - a sequence of roommates, e.g. $[3, 1, 4, 5, 2, 7, 6, 8]$, which results in the following groups: $[\{3, 1, 4, 5\}, \{2, 7, 6, 8\}]$.

For each individual, we calculate a fitness value, which is the average suitability score between pairs in the genes. The way BERGER, R et al. (2018) [1] calculate a score for each pair is by summing the differences between each feature of the two people and then averaging all the pairs' fitness values to get the total score. So the total score for an individual is

$$s = \frac{1}{n} \sum_i^n s_i, \quad (4)$$

where n is the number of pairs.

¹ Evolutionary algorithm and genetic algorithm are used interchangeably.

² It should be noted that an individual in our evolutionary environment and a person in our dataset are a different concepts. The former refers to a creature, expressed as a sequence of genes.

The different features are not weighted by correlation with roommate suitability, hence the suggestion in their future work section that hand assigned weights are introduced for calculating suitability more accurately. We circumvent this approach and the issues it arises by using our neural network model as a suitability score approximator. The non-linear nature of neural networks learns complex relationships between the different features, as opposed to hand-assigned linear weights.

Case $r=2$ If we need to match pairs of roommates in a list, we form inputs by indexing the database to get their features, then get outputs by running our trained neural network (3), and finally, use (4) to calculate total fitness. Here

$$s_i = o, \quad (5)$$

where s_i is a pair of people's score, because our trained neural network is a suitability score approximator.

Case $r>2$ The case is different if we need to match more than two people per shared space. In such a case, we propose that the suitability score be calculated by forming all combinations of people in the group, running each pair input on the neural network and averaging the score

$$s_i = \frac{1}{n} \sum_j^n s_{p_j} \quad (6)$$

to get s_i - the suitability score of one group in the individual. s_{p_j} is a combination pair within the group. This way, we will have quantified the relationships between all group members. Finally, we use (4) to get the individual's total fitness.

If we look at the above given example $[\{3, 1, 4, 5\}, \{2, 7, 6, 8\}]$, we need to calculate the suitability score for the quadruplets to calculate the average fitness value. For the first quadruplet (3, 1, 4, 5) we form all the combinations of two people, i.e 3-1, 3-4, 3-5, 1-4, etc. Each combination we run through the neural network and get a suitability score. We then average them to get the quadruplet's suitability score. The same is done for all quadruplets in the individual's genes. Then they are averaged to finally get the fitness value of an individual. The highest fitness individuals are more likely to pass on their genes to the next generation.

Environment We declare N the initial number of individuals in our environment.

Mutation To create the next generation of individuals a mutation operation is introduced. It works by swapping two genes of an individual, thus creating a

slightly altered individual for the next generation. A parameter α is introduced to control the chance of mutation within an individual. We set our $\alpha = 0.025$. This means that the probability that a swap of two genes occurs is 2.5%. As one swap includes two genes, the probability of a gene being included in a swap operation is 2α (5%).

Elitist selection The best b individuals are selected to be carried over to the next generation as they are (without mutation).

Roulette selection Roulette selection is used to select the individuals from the current generation that should be mutated and carried over to the next one. It simply assigns probabilities to each individual proportional to its fitness. The higher the fitness, relative to others, the higher chance of selection. $N - b$ number of selections are made, as we have b spots reserved for the best-performing individuals of our environment.

When the genetic algorithm has run for some iterations, the individuals are fit enough to be considered a solution to the stable matching problem, thus solving our residence roommate assignment task. The accuracy of our solution depends on the accuracy of our neural network approximator.

6 Validation

6.1 Hypothesis

A deep learning system, which takes into account a person’s preferences, habits and personality traits, yields more suitable roommate matches than a system, which doesn’t take this information into account. Suitability is defined as a measure of how happy two roommates are with each other and it is evaluated based on a Likert-style questionnaire.

6.2 Experimental and Control Condition

Our subjects are a group, ranging from two thousand to four thousand students (the number of freshman students in a given year, admitted to VU Amsterdam). They characterize as male and female students, aged between 17 and 30. Of them, half are assigned to the experimental group at random and the other half - to the control group. The experimental group consists of students, who are presented with our app to assign a roommate to them. The control group consists of students, who apply to a common housing agency to get a roommate. Both groups are asked to contact us, if their housing status changes, i.e they decide to move out. What concerns us is why someone’s housing status changed. Was it because they did not like their roommate or some other reason, not related to their roommate? If not, we look at how much time the two subjects have lived together. If it is long enough for both of them to have a certain opinion on their experience with one another, we hand them an evaluating survey.

6.3 External Validity

Users' happiness with their assigned roommates is assessed through a survey, which is similar to the one for gathering labels for our dataset. It uses Likert-style questions:

Table 2. Table showing the questions in our survey for external validation.

Questions
How happy were you with your roommate?
How much did your roommate hinder your work performance?
How much did your roommate hinder your well-being?
How much were you able to collaborate with your roommate?
How much was privacy an issue?

6.4 Possible Confounding variables

Privacy issues If a participant does not agree with the privacy statement and/or does not give all information asked for, we cannot process the user's data, which is essential for our algorithm. Thus, we cannot include this subject to our experiment. Only complete data can be included.

Non-user-friendly interface If a participant finds the interface of our system to be user unfriendly, it could result in the user not using our app and thus not giving his full information or giving up on the experiment.

Fear of giving criticism If a participant is afraid to criticize their roommate, they would give inaccurate evaluation of their happiness with their roommate, which in turn would decrease the accuracy of our model.

7 Conclusion

Our work enabled us to cover all aspects of the problem of roommate matching and consequently arrive at two core problems, for which we propose systematic solutions. The neural network approach to suitability score approximation circumvents the issue of creating a function that approximates suitability by hand. It learns the non-linear relationships between features in the data automatically, as we have gathered a sufficient dataset of three thousand student pairs for training. The genetic algorithm solution extends the former method's applicability to whole lists of subjects that have to be matched with each other. Since we have created an universal estimator of suitability, we can successfully use it in more complex systems, such as the core of our evolutionary algorithm.

With this study, we aim to advance the practicality of artificial intelligence by solving this very common problem for students. Should the solutions presented in this paper be a success in trials in our university, we could use them throughout Netherlands and even the world.

As technology is being more relied on than ever before and the use of AI in everyday life seemingly being the future of society, it was perhaps time that AI technology was used to improve on the aspects of a university student's life. Ultimately, as long as these methods can be used to make the lives of people easier, then we think such research is viable and applicable.

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